

Project Type: Engineering

Overall Grade: 76

Category Scores

Context Aims Objectives (Weight 2)

Score: 82

Justification: The project provides a clear, relevant context for data exfiltration threats and articulates well-defined aims and objectives. The abstract, introduction, and aims/objectives sections are clearly structured. The aims and objectives are mapped and restated throughout. However, there is some repetition, and the objectives could be ordered for greater clarity.

Transparency: High score was given due to clear linkage between the context, aims, and objectives, referencing rubric requirements for explicit, logical project framing found in multiple report sections.

Reasoning Summary: The report establishes a strong rationale and detailed objectives tightly linked to the project outcomes.

Confidence: 90 – Sections are explicit, structured, and strongly aligned with the rubric. No ambiguity detected.

Literature Review (Weight 2)

Score: 78

Justification: The literature review demonstrates comprehension of recent work on data exfiltration, including attack vectors and detection/prevention shortcomings. Appropriate primary and secondary sources are cited. Some methods and findings from the literature are critically discussed and directly inform methodology. Other relevant recent studies could be further discussed and a more systematic comparison would enhance the section.

Transparency: Score reflects breadth and depth of coverage, with evidence of critical engagement, in line with rubric requirements.

Reasoning Summary: The literature review connects academic research to practical implementation, highlighting gaps and informing the study.

Confidence: 85 – References are cited throughout. Comparative and evaluative elements are evident, suggesting robust scholarship.

Methodological Approach (Weight 1)

Score: 74

Justification: The report evaluates and justifies software development life cycle models, making a logical, evidenced choice of incremental development. Software architecture and the rationale for language/library decisions are well-explained. There could be further elaboration on how the methodological approach affects experimental validity and generalisability.

Transparency: The score reflects explicit mapping of methodological decisions (incremental model, pipeline architecture) to project requirements and constraints.

Reasoning Summary: The approach is justified and fit for purpose, with logical development process selection.

Confidence: 85 – Justification for selected approaches is well-evidenced and linked to project requirements.

Requirements Specification (Weight 3)

Score: 79

Justification: The specification clearly breaks down requirements into high-level needs, functional specifics, and user considerations. Requirements are measurable, and the evaluation cross-references these in detail. However, some requirements for scalability and broader interface options could be more systematically addressed.

Transparency: Score is based on completeness, specificity, and traceability of requirements against rubric points.

Reasoning Summary: Requirements are comprehensive and guide design/implementation effectively.

Confidence: 88 – Requirements are explicitly documented and meaningfully assessed in evaluation.

IT Design Analysis (Weight 3)

Score: 80

Justification: The design section analyses each data exfiltration method, exploring content, header, and meta techniques with clarity and technical depth. Constraints, rationale, and alternatives are discussed. Design choices are mapped from requirements and informed by literature.

Transparency: High score owing to robust analysis and methodical explanation of approaches according to rubric.

Reasoning Summary: IT design analysis is comprehensive, with thoughtful discussion of methodological alternatives and technical trade-offs.

Confidence: 90 – Analysis is precise, well-documented, and covers design-space decisions thoroughly.

Implementation Discussion (Weight 3)

Score: 75

Justification: Each implemented method is described and operational details given. Technical hurdles (e.g., MAC address and video steganography limitations) are openly discussed. The program is modular, extensible, and tested with various data types. A slightly deeper dive into code structure, maintainability, and cross-platform issues would enhance this section.

Transparency: Score reflects clear execution and explicit reporting of implementation choices per rubric guidance.

Reasoning Summary: Implementation is robust, with practical challenges managed and documented.

Confidence: 87 – Logical, clearly described implementations with directly evidenced report content.

Verification Validation (Weight 1)

Score: 67

Justification: Verification/testing is described methodically, including logical tests, user input validation, and use of test data. Validation, via result hashes, is transparent. However, further formalisation of test cases, broader usability feedback, and a more systematic validation approach would increase rigour.

Transparency: Score reflects effective but somewhat informal validation/verification steps compared to rubric ideals.

Reasoning Summary: Verification addresses logic and correctness but could be more exhaustive.

Confidence: 78 – Evidence is clear, though the approach is practical rather than formally rigorous.

Evaluation Against Requirements (Weight 2)

Score: 75

Justification: Evaluation systematically considers project requirements, examining each method's success and limitations. Testing against Snort IDS, discussion of detection/alerting, and development of custom rules demonstrate critical evaluation and insight. Broader consideration of scalability and performance would improve this further.

Transparency: The score follows an explicit, requirements-referenced evaluation aligning with rubric.

Reasoning Summary: Effective coverage of requirement fulfilment and limitations is demonstrated.

Confidence: 83 – Evaluation traces requirements and provides supporting evidence for conclusions.

Project Planning Management (Weight 1)

Score: 73

Justification: There is a clear timeline, well-documented Gantt-like plan, risk analysis, and strategies for version control and backups. The response to the COVID-19 disruption is included. Some execution details (milestone tracking, project resourcing) could be expanded.

Transparency: Score reflects adherence to rubric focus on planning, change management, and risk documentation.

Reasoning Summary: Project planning is thorough, with meaningful risk/mitigation consideration.

Confidence: 80 – All major elements of management are covered with direct evidence.

Solution Attributes (Weight 5)

Score: 76

Justification: The solution is innovative, covers multiple exfiltration methods, is modular, and demonstrates technical sophistication. It is well-adapted to security testing and prevention, supporting extensibility. Certain implementation gaps (e.g., video steganography, MAC, and cross-provider email support) are acknowledged and limits noted.

Transparency: Score reflects completeness, innovation, and impact of the solution according to rubric.

Reasoning Summary: The project shows technical depth and practical value, with evident innovation.

Confidence: 85 – The solution's breadth and efficacy are substantiated by detailed descriptions.

Summary Conclusions Recommendations (Weight 2)

Score: 73

Justification: The conclusions summarize outcomes, issues, and potential future work. Recommendations for technical improvement and further study are explicitly made. More detailed reflections on impacts, generalizability, and lessons learnt would enhance this section.

Transparency: Score reflects thorough, though not exceptionally deep, synthesis and forward-looking perspective.

Reasoning Summary: Relevant conclusions and practical recommendations cap the project.

Confidence: 80 – Supported by clear, direct references to previous findings and future work.

Structure Presentation (Weight 2)

Score: 77

Justification: The report is well-structured, follows logical flow, and employs figures and tables to aid understanding. Minor editorial issues and some redundancy are present, but overall readability is strong.

Transparency: Score is based on effective organisation and clear presentation aligned with rubric.

Reasoning Summary: Presentation is professional, facilitating reader comprehension.

Confidence: 85 – Clarity and structure are evident throughout the document.

Overall Understanding Reflection (Weight 3)

Score: 75

Justification: The project shows a strong grasp of technical, practical, and security implications of data exfiltration. Limitations are openly discussed and reflected upon. Ethical and societal implications are considered. Reflection on learning and personal development could be expanded.

Transparency: Score reflects integration of understanding and critical reflective elements per rubric.

Reasoning Summary: Demonstrated understanding, self-awareness, and insight are clear.

Confidence: 80 – Understanding and reflection are supported by project documentation and critical discussion.

Strengths

The project demonstrates strong technical innovation across a diverse set of data exfiltration methods, supported by comprehensive requirements, robust literature context, and thoughtful evaluation. The report is well-organized, critical, and contextually aware with clear alignment from aims to results.

Weaknesses

There are some implementation gaps in a few methods (e.g., video steganography, MAC address method beyond local networks). Testing and validation could be more formalized, and deeper reflection on learning, scalability, and generalizability would further enhance the work. Minor editorial and redundancy issues are present.

Rubric Alignment

Summary: The report consistently meets or exceeds the majority of rubric criteria across engineering design, implementation, evaluation, and reflection. All core areas are addressed with appropriate depth and transparency, though there are minor shortfalls in validation, scalability discussion, and reflective synthesis.

Threshold Checks

Missing Criteria: No major rubric elements are missing. Minor gaps exist in formal validation, scalability, and deeper reflective commentary.

Overall Confidence: 86 – Category-level confidences are high in all sections, with direct evidence supporting judgements. Minor uncertainties arise from areas of limited depth or breadth in evaluation and reflection.